Lav Kush Vidhya Vihar H. S. School

Half-Yearly Examination: 2025-2026

Class – 11th

Time : 3Hrs. Subject – Maths M.M. : 100

Q.1. Choose the correct option- 06

1) The radian measure of 240 is
   a) $[4\frac{\pi}{3}]^c$ b) $[2\frac{\pi}{3}]^c$ c) $5\frac{\pi}{2}$c d) $3\frac{\pi}{2}$c
2) The solution of 2($2x$ +3) – 10< 6 (x-2) is
   a) (Infinity, u) b) (4, infinity) c) 0, infinity) d) (infinity ,0)
3) The value of i^9 + i^{19} is
   a) 3 b) 1 c) 2 d) 0
4) The range of real function f(x) = -I xl is given by
   a) (0,∞) b) [0,0] c) (-∞,0] d) (0,∞)
5) The function T Which make temperature in degree Celsius Into temperature in degree Fahrenheit is defined by T(c) = $\frac{9C}{5}$ + 32 then t(10) is given by
   a) 77 B) 88 C) 50 D) 10
6) The distance between the pair of paints (-1,3 ,-4)and (1, -3, 4)
   a) $2\sqrt{26}$ b) $2\sqrt{13}$ c) 3 $\sqrt{26}$ d) $3\sqrt{13}$

Q.2. Fill in the blanks :-

1) Value of ∟0 _____
2) The value of sin 150 _____
3) If x is a real number then solution of inequality 3x+1< 5x+7 is _____
4) If f(x)=1-sinx then value of f($\frac{\pi}{2}$)=_____.
5) Conjugate of -4+5i=_____.
6) Distance between the points (1,2,3) and (1,3,-2) is _____.

Q.3. Match the column: **(6)**

'A'	**'B'**
1. $Sin\ (\frac{3\pi}{2} - x)$	(a) $Cos^2x - Sin^2x$
2. $cos2x$	(b) 1 $\frac{2tanx}{1+tan^2x}$
3. $tan2x$	(c) $\frac{1-tan\theta}{1+tan\theta}$
4. $tan\ \left(\frac{\pi}{4} + \theta\right)$	(d) $\frac{1+tan\theta}{1-tan\theta}$
5. $tan\ \left(\frac{\pi}{4} - \theta\right)$	(e) $-cosx$
6. $Cos^2x + Sin^2x$	(f) $\frac{2tanx}{1-tan^2x}$
	(h) $\frac{1-tan\theta}{1+tan\theta}$

Q.4. Write the answer on one word / sentence. 07

1. Find the value of i^{23}
2. If f (x) = $\frac{x^2}{1-x^2}$ find the value of f ($\sin\theta$)
3. If A= { 2,4,6} then write all the subject of set A
4. Coordinate (-2,3,4) lie in octant ?
5. Median 3,9,5,3,12,10,18,4,7,19,21.
6. How many 4 digit number all them with no digit repeated ?
7. 3Median =_____

Section –B

Q.1. Write interval in set builder form [6,12] 02

Or

If A={1,2,3,4} B={3,4,5,6} D={7,8,9,10}

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Q.2. If A={1,2} and B={3,4} write A×B how many subsets will A×B have list them. 02

Or

If f(x)=3x-5 write down the value of

1. f(-2)| 2. F(0)

Q.3. Solve $\lim_{x \to \pi} \frac{Sin}{\pi} \frac{(\pi-x)}{(\pi-x)}$ 02

Or

Find the derivative of $\frac{x+1}{x-1}$ w.r.t. 'x'

Q.4. Explain in the form of $a + ib$ 02

$\frac{1}{2} + \frac{2}{5}i) - (4 \leftarrow \frac{5}{2}\ i)$

Or

Find multiplicative inverse of 4-3i.

Q.5. Solve the inequality $5x - 3 < 7$ when 02

i) x is an integer

ii) x is a Real number

Or

Show the inequality on number line $5x - 3 > 3x - 5$

Q.6. How many 3digit number can be formed by writing the digit 1 to 9 if no digit repeated?

Or 02

Find n if $np_3 : np_4 = 1:9$

Q.7. Find the coordinates of the point which divides the line segment joining the point (1,-2,3) and (3,4,-5) in ratio 2:3 . 02

Or

Find the ratio in which the yz plane divides the line segment joining the point (1, -2,3) and (3,4,-5) in ratio (2:3)

Q.8. Find the value of other five trigonometric function if cot x = $\frac{3}{4}$ lies in third quadrant. 02

Or

Prove that $\frac{\text{Sin}\, x+Sin3x}{\text{Cos}\, x+Cos3x}$ $=tan2x$

Q.9. Solve the equation 02

$X^2+3x+5=0$

Or

Solve $(1-i)^4$

Q.10. If A= { 3,6,9,12,15,18,21 } B={4,8,12,16,20} c={2,4,6,8,10} then find 02

i) A-C ii) C-A

Or

if U={1,2,3,4,5,6,7,8,9} A={2,4,6,8} b= {2,3,5,7} then verify (AUB) = A' NB'

Q.11. Let A={1,2,3,-------14} define a relation R prime A by A R= {x,y : 3x-4=0} where x,y |(-4) write down its domain, codomain, and range. 03

Or

Write down the relation R= {x,x^3} x is a prime number less than 10} in roster from .

Q.12. Prove that $\frac{Cos4x+Cos3x+Cos2x}{sin4x+sin3x+sin2x}$ =Cot3x 03

Or

Prove that $cotx\ cot2x - cot2x\ cot3x - cot3x\ cotx = 1$

Q.13. For any two complex number z_1 and z_2 prove that 03

Re (z_1z_2) = Re(z_1) Re(z_2) – IM (z_1) IM (z_2)

Find the modulus of $\frac{1+i}{1-i} - \frac{1-i}{1+i}$

Q.14. Find the derivative of $Cosx$ from first principal. 03

Or

Find the derivative of $\frac{ax+b}{cx+d}$ w.r.t. $x\ \ o$

Q.15. Find the equation of sell of the points P such that Its distances from the points A (3,4,-5) and B (-2,1,4) are equal. 03

Or

Show that the point A (1,2,3) B (-1,-2,-1) C (2,3,2) And D (4,7,6) are the vertices of a Parallelogram ABCD But it is not rectangle.

Q.16. Prove that $2\text{Sin}^2\ \frac{\pi}{6} +$, $\text{Cosec}^2\ \frac{7\pi}{6}\ \text{Cos}^2\ \frac{\pi}{3} = \frac{3}{2}$ 04

Or

$tan\ (\frac{\pi}{4} + x)$ = $(1+\tan x)^2$

$\tan\ (\frac{\pi}{4} - x)$ = $(1 - tanx)^2$

Q.17 Find the mean deviation about median for the following data: 04

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of girls	6	8	14	16	4	2

OR

Find the mean deviation about mean.

Marks	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of girls	2	3	8	14	8	3	2

Q.18 In how many ways can a student choose a programme of 5 courses. If 9 courses are available and 2 specific courses are compulsory for every student. 04

OR

How many 3 digit numbers can be formed from the digits 1,2,3,4 and 5 assuming that:

1) Repetition of the digit is allowed?
2) Repetition of the digit is not allowed?

Q.19 In a group of 70 people, 37 like coffee, 52 like tea and each people like at least one of the two drinks. How many people like both coffee and tea. 04

OR

In a group of 65 people, 40 like cricket,10 like both cricket and tennis only and not cricket? How many like tennis?